Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/27/2023
PI Name:	Bhattacharjee, Subrata Ph.D.		
Project Title:	Residence Time Driven Flame Spread: The Final Phase		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92182-0001	**Congressional District:**	53
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	06/01/2021	**End Date:**	05/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Miller, Fletcher Ph.D. (San Diego State University) Paolini, Christopher Ph.D. (San Diego State University) Takahashi, Shuhei Ph.D. (Gifu University, Japan)		
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Task Description:	The continuing work builds upon our ground and space based studies so far – theoretical, computational, and experimental – to investigate steady and unsteady flame propagation over solid fuels in a microgravity environment. One of the central hypotheses of the Residence Time Driven Flame Spread (RTDFS) project is that steady flame spread over solid fuels cannot be sustained in a microgravity environment, even at high oxygen environment, if the fuel is thicker than a critical thickness. At lower oxygen level, a similar situation arises when the opposing flow velocity is below a critical velocity. A simplified theory based on scale analysis has produced closed-form formulas in terms of known parameters for these critical quantities. A comprehensive computational model that includes gas and surface radiation, finite-rate chemistry for pyrolysis, and combustion support these results. Moreover, computations reveal that the asymmetry between the thermal and species fields introduced by radiative losses is responsible for a slowly moving flame to be surrounded by a vitiated atmosphere leading to extinguishment. Earlier space based experiments showed a slowing flame over thick poly(methyl methacrylate)/PMMA slabs consistent with our findings. Also, experiments in the Burning and Suppression of Solids-II (BASS-II) project, conducted in 2019, has produced a wealth of data at a relatively low oxygen level that are consistent with our prediction. However, no space based experiments have ever been performed to test the hypothesis that steady flame spread over fuels exceeding a certain thickness is unsustainable even at high oxygen level. The proposed space based experiments in the Solid Fuel Ignition and Extinction (SoFIE) project are intended to establish the critical fuel thickness, critical flow velocity, and experimentally establish the mechanism of flame extinguishment. The current extension proposal is to continue our ground based effort for the successful execution of the flight experiments and processing the data from the flight experiments to test the proposed hypotheses. Some of this work includes: (a) Refining the numerical model to create grid and domain independent simulation of opposed-flow flame spread in four configurations: microgravity, thermal, kinetic, and downward. (b) Parametric studies involving fuel thickness, oxygen level, pressure, opposing-flow velocity, and ambient pressure. (c) Improve the pyrolysis kinetics model. (d) Evaluate the effect of solid-phase conduction in flame spread. (e) Develop expressions for flame length in different regimes. (f) Theoretical work to establish a comprehensive set of non-dimensional numbers, supported by computation, to delineate different regimes of flame spread. (g) Further develop the MATLAB based Flame Image Analysis Tool (FIAT) application to be able to analyze videos from the SoFIE experiments. (h) Conduct experiments in the Flame Tunnel and Flame Stabilizer at San Diego State University (SDSU) to test our numerical model. (i) Modify the Ignition Apparatus at SDSU to conduct ignition experiments to complement ignition delay data from all SoFIE experiments to develop an ignition model for solid fuel in a microgravity environment. (j) Process the data obtained from the Space experiments to test the hypothesis of the original proposal. (k) Further develop the educational outreach web portal, www.themofluids.net, currently used by over 3000 educators. (l) Publish the research result in renowned combustion journals and conferences. Establishing a critical non-dimensional parameter that leads to self-extinguishing flame in a low-gravity environment has a direct implication to fire safety in spacecrafts. It can produce a critical thickness, a critical velocity, a critical pressure, or oxygen level that makes a thin combustible material inherently fire-safe. Validating the mechanism of such self-extinguishing flames – that a flame chokes itself due to the asymmetry of its species and thermal fields, established numerically – from the RTDFS experiments will be a significant scientific discovery. NOTE: Project continues as second phase of "Residence Time Driven Flame Spread Over Solid Fuels--NNX15AG11G".
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research has three components: We have built a novel experimental facility called Flame Tower at SDSU, where ground based experiments on flame spread are being conducted; Theoretical and computational work that explore the similarity and differences between the mechanisms flame spread in a zero gravity space environment and on Earth; Design and propose a space based experiment to establish extinction mechanism of flames. The data that we are acquiring in our ground based work will provide the research community with a comprehensive set of results for testing different theories of flame spread and solid fuel ignition in a normal gravity environment. The space-based experiments will test our hypothesis that a flame spread over a flat solid fuel is unstainable if the fuel thickness exceeds a critical thickness. This will have implications in designing fire safe habitat in space. The thermodynamic portal, www.thermofluids.net, is used by a huge number of students and educators around the world.
Task Progress:	Formulas for calculating the flame spread rate (FSR) for opposed-flow flame spread over a thin or thick flat fuel are well known. We extended these formulas through a theoretical analysis to predict the FSR for cylindrical fuels. The theoretical predication was compared to the experiments performed in our laboratory. This work led to a publication in the Proceeding of the 39th Symposium of Combustion. Our computational work on fuel pyrolysis, done with collaboration from Dr. Michael Delichatsios, led to a publication in the Proceeding of the 39th Symposium of Combustion. In this work we examined how the pyrolysis temperature depends on the environmental parameters and properties of the fuel and oxidizer. Our collaborative experimental work with Prof. Tsuney Matsuoka of Toyohashi University of Technology led to another publication in the Proceeding of the 39th Symposium of Combustion. In this work we examined how flame spread over solid fuels is enhanced by the presence of edges. Work on how pyrolysis affects flame shape and heat flux prediction in an ignition device led to presentations in the 51st ICES meeting and 38th ASGSR meetings.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Bhattacharjee, S., Delzeit, T., "Opposed-Flow Flame Spread over Cylindrical Fuels: Spread Rate Formulas and Experimental Verification" Proc. Combust. Inst., https://doi.org/10.1016/j.proci.2022.07.243 , Oct-2022

Articles in Peer-reviewed Journals	Bhattacharjee, S., Delichatsios, M., "Predicting the Pyrolysis Temperature for Thermally Thin Fuels in Opposed-Flow Flame Spread in the Thermal Regime" Proc. Combust. Inst., https://doi.org/10.1016/j.proci.2022.07.246 , Oct-2022
Articles in Peer-reviewed Journals	Matsuoka, T., Chiba, Y., Okuno, S., Torikai, H., Bhattacharjee, S., " Effect of internal angle on opposed flame spread over a folded plate" Proc. Combust. Inst., https://doi.org/10.1016/j.proci.2022.07.153 , Oct-2022
Dissertations and Theses	Kilian Schütte "An Experimental Study of Flame Spread Over Solid Fuels with Metallic Core: The Role of Longitudinal Conduction" Universität der Bundeswehr in Munich and San Diego State University , Sep-2022
Dissertations and Theses	Laurenz Bissada "The Influence of Dripping on the Downward Spread Rate of PMMA Fuels and Burn Characteristics" Universität der Bundeswehr in Munich and San Diego State University , Sep-2022
Dissertations and Theses	Leander Randhofer "Evaluating Smoke-Time Delay and Steady-State Temperature of Coated Ignition Wire," Universität der Bundeswehr in Munich and San Diego State University , Sep-2022
Papers from Meeting Proceedings	Okuno, S, Matsuoka, T., Takuya, Y., Bhattacharjee, S., Torikai, H, Nakamura, Y. "Effect of thickness on flame spread over a V-shaped plate" The 60th Symposium (Japanese) on Combustion Paper #234, C122 , Oct-2022
Papers from Meeting Proceedings	Bhattacharjee S., Vera, E., Casebier, N., Creating a Prescribed Heat Flux in an Ignition Experiment "Creating a Prescribed Heat Flux in an Ignition Experiment" 38th Meeting ASGSR, Houston Paper #182, , Nov-2022
Papers from Meeting Proceedings	Bhattacharjee S., Delichatsios, M., "Competition Between Pyrolysis Kinetics and Surface Radiation in Opposed-Flow Flame Spread in a Microgravity Environment" 51st ICES Meeting, St. Paul paper #128 , Jul-2022
Papers from Meeting Proceedings	Bhattacharjee, S. "Expressions for Flame Spread Rate over Cylindrical Fuels I Opposed-Flow Configuration Using a Scale Modeling Approach" 9th ISSM, Napoli, Italy Paper # 27 , Mar-2022